Pembroke Hill Invitational

12/6/2025

Machines Test Packet

Division B

Wherever necessary, assume g=9.8 m/s2 unless otherwise stated. Show work on your scratch paper.

1. **True or False:** [1 pt]
   In a real-world machine (where friction exists), the Actual Mechanical Advantage (AMA) can never be greater than the Ideal Mechanical Advantage (IMA)

2. Which class of lever is represented by a human forearm lifting a weight (bicep provides effort, elbow is fulcrum, hand holds load)? [1 pt]

3. If you increase the length of a wedge but keep its thickness constant, what happens to its Ideal Mechanical Advantage (IMA)? [1 pt]
   A) Increases
   B) Decreases
   C) Stays the same
   D) Becomes zero

4. A screw is technically a variation of which other simple machine wrapped around a cylinder? [1 pt]

5. True or False:
   A single fixed pulley changes the magnitude of the input force required to lift a load (assume ideal conditions). [1 pt]

6. Which quantity is always conserved in an ideal simple machine, but not in a real machine? [1 pt]
   A) Force
   B) Work
   C) Power
   D) Velocity

7. A standard manual stapler (when stapling a paper) acts primarily as which class of lever? [1 pt]

8. In a wheel and axle system, if the effort is applied to the **axle** rather than the wheel, the mechanical advantage will be: [1 pt]
   A) Greater than 1
   B) Less than 1
   C) Equal to 1
   D) Infinite

9. A screwdriver with a handle diameter of 4 cm is used to turn a screw with a pitch of 2 mm. What is the IMA of this system? ($\pi \approx 3.14$) [2 pts]

10. A doorknob has a radius of 4 cm, and the internal spindle (axle) has a radius of 1 cm. If it takes 20 N of friction to retract the latch, how much force must you apply to the knob to open it? [2 pts]

11. If you are using a block and tackle system with 4 supporting strands of rope. What is the IMA of this system and what type of Machine is this? [2 pts]

12. How much work have you done is you pushed a box 10 meters across a floor with a force of 50 N. [1 pt]

13. A driver gear has 8 teeth and turns at 200 RPM. It drives a follower gear with 40 teeth. What is the RPM of the follower gear? [1 pt]

14. You have a lever pushing a block up an inclined plane. The lever has an IMA of 3, and the inclined plane has an IMA of 4. What is the total IMA of this compound machine system? [1 pt]

15. A worker uses a pulley system with an efficiency of 80% to lift a 100 kg crate. If the pulley has 5 supporting strands, what actual input force (in Newtons) must the worker apply? (Use g=10 m/s2) [2 pts]

16. A motor with 90% efficiency drives a pulley system with 80% efficiency. What is the total efficiency of the system? [1 pt]

17. Two masses, 5 kg and 15 kg are placed at opposite ends of a massless 4-meter beam. How far from the 5 kg mass must the fulcrum be placed to balance the beam? [2 pts]

18. A wedge with an IMA of 5 is driven into a log. The resistive force of the wood is 2000 N. If it actually takes 600 N of force to drive the wedge, what is the force lost to friction? [2 pts]

19. A ladder leans against a smooth wall. The ground applies both a normal force and a friction force. Which simple machine principle best describes the relationship between the ladder's angle and the likelihood of slipping (related to the coefficient of friction)? [2 pts]

20. Which Greek mathematician is famously quoted as saying, "Give me a place to stand, and I shall move the earth," referring to the principle of the lever? [1 pt]

21. A compound machine consists of a lever (IMA = 3.0) operating a pulley system (IMA = 2.0). If the entire system is 80% efficient, what is the Actual Mechanical Advantage (AMA) of the compound machine? [1 pt]

A) 4.0 B) 4.8 C) 5.0 D) 6.0

22. Which of the following changes would theoretically DECREASE the Ideal Mechanical Advantage (IMA) of a screw? [1 pt]

A) Increasing the diameter of the screwdriver handle used to turn it
B) Decreasing the distance between the threads (pitch)
C) Increasing the number of threads per inch (TPI)
D) Decreasing the circumference of the screwdriver handle

23. Consider a standard household broom being used to sweep dust. Which of the following best describes the mechanics of this system? [1 pt]

A) Class 1 Lever; it multiplies force
B) Class 2 Lever; it multiplies force
C) Class 3 Lever; it multiplies speed/distance
D) Class 3 Lever; it multiplies force

24. Which of the following statements about static equilibrium is FALSE? [1 pt]
A) The net force acting on the object must be zero
B) The net torque acting on the object must be zero
C) The object must be stationary (velocity = 0)
D) The object can be moving at a constant velocity

25. In a frictionless world, you push a 100 N box up a 5-meter long ramp to a height of 2.5 meters. You then lift the same box vertically 2.5 meters. Which statement is true? [1 pt]
A) You did more work using the ramp
B) You applied more force using the ramp
C) You did the same amount of work in both cases
D) You applied the same amount of force in both cases

26. Which unit is dimensionally equivalent to a Joule (J)? [1 pt]
A) N * s
B) kg * m/s2
C) N * m
D) Watt/s

27. A wedge with a length of 10 cm and a thickness of 2 cm is used to split wood. If a second wedge has the same length but is 4 cm thick, how does its IMA compare to the first wedge? [1 pt]
A) The second wedge has half the IMA
B) The second wedge has double the IMA
C) The second wedge has the same IMA
D) The second wedge has 4 times the IMA

28. You are designing a Class 1 lever to lift a heavy rock. To maximize the Mechanical Advantage, where should the fulcrum be placed? [1 pt]
    A) Exactly in the middle of the lever
    B) As close to the applied effort force as possible
    C) As close to the rock (load) as possible
    D) The position of the fulcrum does not affect the IMA

29. Which simple machine is mathematically most similar to a screw? [1 pt]
    A) A wedge
    B) An inclined plane wrapped around a cylinder
    C) A second-class lever
    D) fixed pulley

30. If a machine has an Actual Mechanical Advantage (AMA) of 1.0, what does this imply? [1 pt]
    A) The machine is 100% efficient
    B) The output force is exactly equal to the input force
    C) The output distance is greater than the input distance
    D) The system has no friction

31. A force of 20 N is applied to a wrench. Which angle of application (relative to the handle) generates the maximum torque? [1 pt]
    A) 0 degrees (parallel to the handle)
    B) 30 degrees
    C) 45 degrees
    D) 90 degrees (perpendicular to the handle)

32. Which of the following best defines "Power" in the context of simple machines? [1 pt]

    A) The capacity to do work
    B) The rate at which work is done
    C) The total force multiplied by distance
    D) The efficiency of the energy transfer

33. A ramp is used to load furniture into a truck. If the ramp is replaced with one that is twice as long (reaching the same height), what happens to the force required to push the furniture (assuming negligible friction)? [1 pt]

A) The force required doubles
B) The force required is cut in half
C) The force required stays the same
D) The force required decreases by a factor of 4

34. What is the primary mechanical purpose of a single fixed pulley? [1 pt]

A) To reduce the effort force required to lift a load
B) To increase the speed of the load
C) To change the direction of the applied force
D) To increase the work output

35. Which of the following scenarios results in zero work being done on an object? [1 pt]

A) Lifting a 50 kg weight 1 meter off the ground
B) Pushing a stationary wall with 100 N of force for 5 minutes
C) Pushing a box across a floor with friction
D) Lowering a weight slowly to the ground

36. If a simple machine requires 50 Joules of input energy to produce 40 Joules of useful output work, what is the efficiency? [2 pt]

37. Two gears are meshed. Gear A has 10 teeth and turns at 100 RPM. Gear B has 50 teeth. How fast is Gear B turning? [1 pt]

A) 20 RPM
B) 50 RPM
C) 200 RPM
D) 500 RPM